

DBMS Assignment No 5

Smart Gravience Handling

PROCEDURE

1. Insert Issue & Calling :

```
[mysql> use pccoe;
Database changed
mysql> DELIMITER $$
mysql>
mysql> CREATE PROCEDURE AddIssue(
->     IN p_title VARCHAR(255),
->     IN p_desc TEXT,
->     IN p_area VARCHAR(100),
->     IN p_severity VARCHAR(10),
->     IN p_citizen_id INT
-> )
-> BEGIN
->     INSERT INTO Issue(title, description, area, severity, citizen_id)
->     VALUES(p_title, p_desc, p_area, p_severity, p_citizen_id);
-> END $$
Query OK, 0 rows affected (0.004 sec)

mysql>
[mysql> DELIMITER ;
[mysql> CALL AddIssue('Garbage issue', 'Overflow garbage', 'Nigdi', 'High', 1);
Query OK, 1 row affected (0.009 sec)
```

Output :

```
[mysql> SELECT * FROM Issue;
```

issue_id	title	description	area	severity	priority_score	status	report_date	resolution_date	citizen_id
1	Pothole near highway	Large pothole causing accidents	Akurdi	High	0	Open	2026-03-31 22:35:15	NULL	1
4	Street light not working	Dark area at night	Chinchwad	Low	0	Open	2026-03-31 22:35:15	NULL	4
5	Check trigger	Testing	Akurdi	High	90	Open	2026-03-31 22:45:46	NULL	1
7	Garbage issue	Overflow garbage	Nigdi	High	90	Open	2026-04-13 21:57:00	NULL	1

```
4 rows in set (0.001 sec)
```

2.Update Issue Status & Calling :

```
mysql> DELIMITER $$
mysql>
mysql> CREATE PROCEDURE UpdateIssueStatus(
    ->     IN p_id INT,
    ->     IN new_status VARCHAR(20)
    -> )
    -> BEGIN
    ->     UPDATE Issue
    ->     SET status = new_status
    ->     WHERE issue_id = p_id;
    -> END $$
Query OK, 0 rows affected (0.002 sec)

mysql>
[mysql> DELIMITER ;
[mysql> CALL UpdateIssueStatus(1, 'Resolved');
Query OK, 1 row affected (0.002 sec)
```

Output :

```
mysql> SELECT * FROM Issue;
```

issue_id	title	description	area	severity	priority_score	status	report_date	resolution_date	citizen_id
1	Pothole near highway	Large pothole causing accidents	Akurdi	High	0	Resolved	2026-03-31 22:35:15	NULL	1
4	Street light not working	Dark area at night	Chinchwad	Low	0	Open	2026-03-31 22:35:15	NULL	4
5	Check trigger	Testing	Akurdi	High	90	Open	2026-03-31 22:45:46	NULL	1
7	Garbage issue	Overflow garbage	Nigdi	High	90	Open	2026-04-13 21:57:00	NULL	1

```
4 rows in set (0.001 sec)
```

3.Delete Issue and Calling :

```
mysql> DELIMITER $$
mysql>
mysql> CREATE PROCEDURE DeleteIssue(
    ->     IN p_id INT
    -> )
    -> BEGIN
    ->     DELETE FROM Issue WHERE issue_id = p_id;
    -> END $$
Query OK, 0 rows affected (0.004 sec)
```

Output :

```
[mysql> CALL DeleteIssue(4);  
Query OK, 1 row affected (0.004 sec)
```

4.Show all Issue and Calling :

```
mysql> DELIMITER $$  
mysql>  
mysql> CREATE PROCEDURE ShowIssues()  
-> BEGIN  
->     SELECT * FROM Issue;  
-> END $$  
ERROR 1304 (42000): PROCEDURE ShowIssues already exists  
mysql>  
[mysql> DELIMITER ;  
[mysql> CALL ShowIssues();
```

Output :

issue_id	title	description	area	severity	priority_score	status	report_date	resolution_date	citizen_id
1	Pothole near highway	Large pothole causing accidents	Akurdi	High	0	Resolved	2026-03-31 22:35:15	NULL	1
5	Check trigger	Testing	Akurdi	High	90	Open	2026-03-31 22:45:46	NULL	1
7	Garbage issue	Overflow garbage	Nigdi	High	90	Open	2026-04-13 21:57:00	NULL	1

3 rows in set (0.001 sec)

Query OK, 0 rows affected (0.001 sec)

FUNCTIONS

1. Get Priority Score :

```
mysql> DELIMITER $$
mysql>
mysql> CREATE FUNCTION GetPriority(p_id INT)
  -> RETURNS INT
  -> DETERMINISTIC
  -> BEGIN
  ->     DECLARE p INT;
  ->
  ->     SELECT priority_score INTO p
  ->     FROM Issue
  ->     WHERE issue_id = p_id;
  ->
  ->     RETURN p;
  -> END $$
Query OK, 0 rows affected (0.003 sec)

mysql>
[mysql> DELIMITER ;
[mysql> SELECT GetPriority(1);
+-----+
| GetPriority(1) |
+-----+
|              0 |
+-----+
1 row in set (0.003 sec)
```

2. Count Total Issue :

```
mysql> DELIMITER $$
mysql>
mysql> CREATE FUNCTION TotalIssues()
  -> RETURNS INT
  -> DETERMINISTIC
  -> BEGIN
  ->     DECLARE total INT;
  ->
  ->     SELECT COUNT(*) INTO total FROM Issue;
  ->
  ->     RETURN total;
  -> END $$
Query OK, 0 rows affected (0.004 sec)

mysql>
[mysql> DELIMITER ;
[mysql> SELECT TotalIssues();
+-----+
| TotalIssues() |
+-----+
|              3 |
+-----+
1 row in set (0.002 sec)
```

3.Count Issue By Area :

```
mysql> DELIMITER $$
mysql>
mysql> CREATE FUNCTION CountByArea(p_area VARCHAR(100))
  -> RETURNS INT
  -> DETERMINISTIC
  -> BEGIN
  ->     DECLARE total INT;
  ->
  ->     SELECT COUNT(*) INTO total
  ->     FROM Issue
  ->     WHERE area = p_area;
  ->
  ->     RETURN total;
  -> END $$
```

Query OK, 0 rows affected (0.003 sec)

```
mysql>
mysql> DELIMITER ;
mysql> SELECT CountByArea('Nigdi');
```

CountByArea('Nigdi')
1

1 row in set (0.002 sec)

```
mysql> SELECT CountByArea('Akurdi');
```

CountByArea('Akurdi')
2

1 row in set (0.001 sec)